DriveFusion IVI: A Conceptual Service-Oriented Architecture for Media, Navigation and Smartphone Projection with a Policy-First Context Layer

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***Abstract*—DriveFusion IVI is a conceptual, service-oriented in-vehicle infotainment (IVI) architecture that integrates media, navigation and smartphone projection through a common IVI core. Beyond these baseline features, it introduces a Context Fusion Engine and a Driver-Distraction Policy Engine so that the presentation of information can adapt to vehicle state and interaction context. A distinctive design decision is that personalization machine learning is deliberately placed downstream of a deterministic safety policy, so that convenience intelligence can never override safety-oriented control logic. The paper presents the layered architecture, an interface catalogue, end-to-end data flows, security and reliability measures, a platform-neutral technology mapping and a validation plan. The work is a conceptual academic design and not a production safety design or certification claim.**

***Index Terms*—In-vehicle infotainment, software-defined vehicle, service-oriented architecture, smartphone projection, driver distraction, context fusion, automotive HMI.**

I. INTRODUCTION

Modern in-vehicle infotainment systems must combine media playback, navigation and smartphone projection while remaining safe to use during driving. When each application independently decides how it presents content, the driver's interaction with the system can become inconsistent and potentially distracting. This work proposes DriveFusion IVI, a conceptual architecture in which shared context and a deterministic policy layer govern what the driver is allowed to see and do.

The main contributions are: (i) a five-layer conceptual architecture with annotated interfaces and data flows; (ii) an interface catalogue and end-to-end data-flow definitions; (iii) a policy-first context layer that separates convenience intelligence from safety-oriented control; and (iv) a security, reliability and validation approach suitable for later simulation.

II. REQUIREMENT INTERPRETATION

The required outcome is a conceptual IVI architecture with media, navigation and projection features, together with annotated interfaces and data flow. The proposed design therefore makes the following elements explicit: inputs, connectivity gateways, core services, intelligent context processing and driver-facing outputs.

III. CONCEPTUAL ARCHITECTURE

The complete architecture and its annotated data flows are shown in Fig. 1 (next page). The design is organized into five layers: user and vehicle inputs, connectivity and edge gateways, IVI core services, an intelligent context layer, and driver-facing and cloud outputs. A driver safety boundary encloses the outputs, and the figure also lists the major data types, key data flows and benefits.

Table I summarizes the major data types exchanged and their formats. The architecture is intended to give a unified media, navigation and projection experience; AI-driven context and personalization; driver-distraction prevention; a secure and connected ecosystem; and scalability for future features such as AR, V2X and ADAS.

TABLE I. Major Data Types and Formats

Data type	Format	Data type	Format
Media metadata	JSON	Vehicle signals	CAN-FD
Playback commands	JSON	GNSS position	NMEA
Route / traffic data	JSON	Audio stream	PCM
Projection control	JSON	HMI events	JSON
Telemetry logs	JSON		

IV. LAYERED ARCHITECTURE

A. Inputs

Inputs comprise driver interaction, steering controls, voice, GNSS, the vehicle CAN-FD bus and the smartphone companion application.

B. Connectivity

USB-C, Bluetooth, Wi-Fi and Ethernet terminate in controlled gateways. The Vehicle Gateway manages vehicle-facing routing, while the Projection Gateway manages smartphone sessions.

C. IVI Core

Media, navigation, projection, audio, HMI/display, vehicle-data abstraction and security/privacy are implemented as separate services, allowing each to evolve independently.

D. Intelligent Context

Context fusion combines vehicle, route, media and projection state. A deterministic policy engine decides which interactions are allowed; the personalization service recommends content without overriding policy.

E. Outputs

Outputs are the center display, instrument cluster/head-up display (HUD), speakers, steering controls and the projection UI.

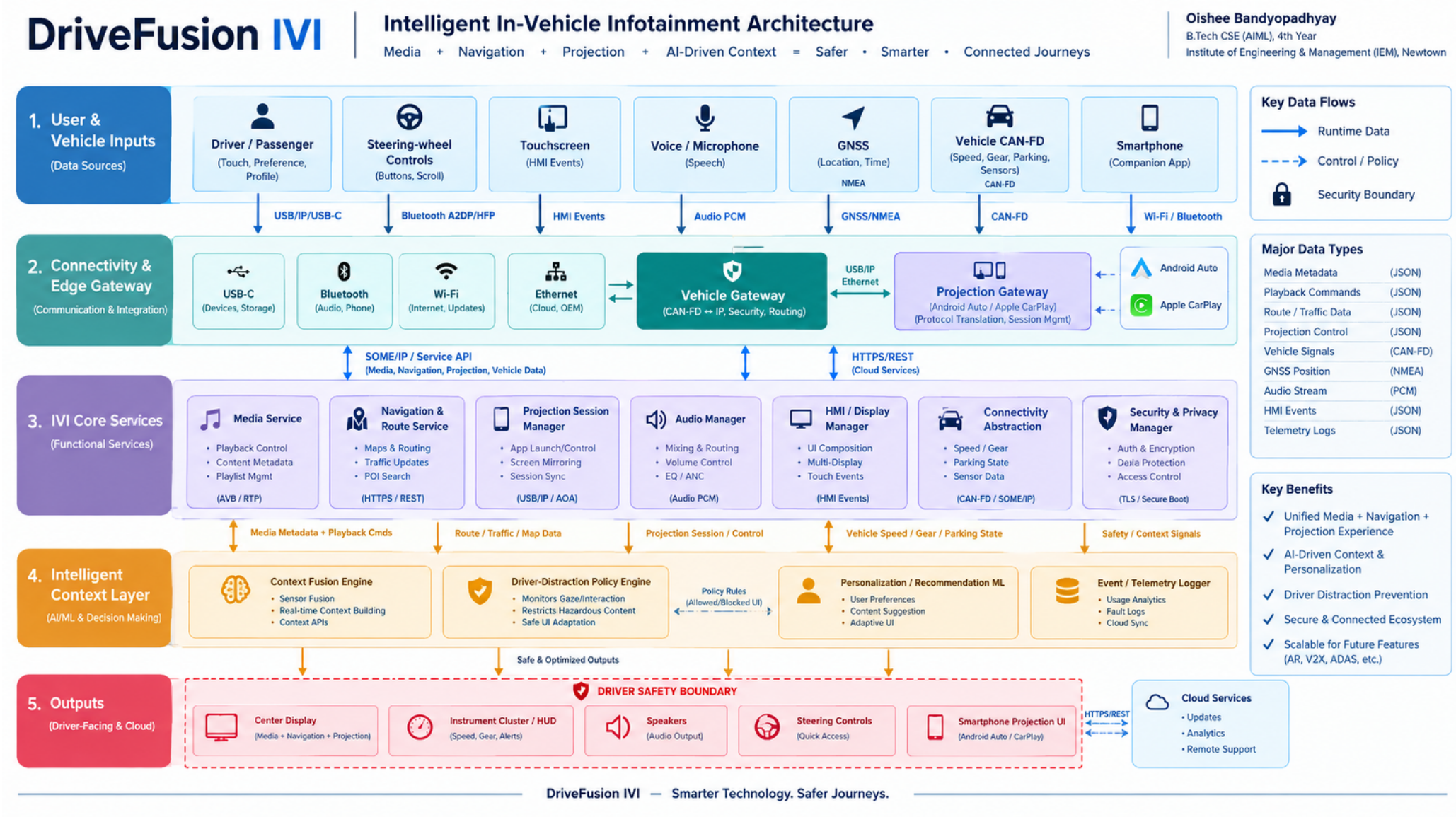


Fig. 1. DriveFusion IVI conceptual architecture and annotated data flows (landscape page for legibility).

V. INTERFACE CATALOGUE

Table II lists the nine principal interfaces. Each interface has an assigned technology and a control mechanism, such as validation, authentication, least-privilege access or a policy gate, so that data contracts and security boundaries are explicit.

TABLE II. Interface Catalogue

ID	Interface	Data	Technology	Control
IF-01	Vehicle Gateway ↔ CAN-FD (vehicle signals)	Speed, gear, parking state, sensors	CAN-FD	Validate / filter
IF-02	Projection Gateway ↔ Smartphone (projection)	Session, UI, controls	USB/IP, Wi-Fi, Bluetooth	Authentication
IF-03	Gateway ↔ IVI Services (service data)	Normalized vehicle properties	SOME/IP / service API	Least privilege
IF-04	Media ↔ Audio (playback)	Play/pause, source, volume, metadata	Local API	Driver-safe controls
IF-05	Navigation ↔ HMI (navigation UI)	Route, maneuver, map state	Service API	Policy gate
IF-06	Projection ↔ HMI (projection UI)	Session UI and controls	Local service/API	Policy gate
IF-07	Context ↔ Policy (context)	Speed, route, media, projection state	Event/API	Deterministic rules
IF-08	Policy ↔ HMI (policy decision)	Allowed / restricted UI	Local API	Safety boundary
IF-09	IVI ↔ Cloud (lifecycle)	Updates, telemetry, analytics	HTTPS / REST	TLS / device identity

VI. END-TO-END DATA FLOWS

Five end-to-end scenarios describe how data traverses the architecture (Table III). The context flow (DF-04) is the basis of the policy-first design because it aggregates vehicle, route, media and projection state before any presentation decision is made.

TABLE III. End-to-End Data Flows

ID	Scenario	Path	Key Data
DF-01	Media	Source → Media → Audio → Speakers	Metadata, commands, audio
DF-02	Navigation	GNSS/Cloud → Navigation → HMI/Cluster/HUD	Position, route, maneuver, traffic
DF-03	Projection	Phone → Projection Gateway → Session Manager → HMI	Session and app content
DF-04	Context	Vehicle + Route + Media + Projection → Fusion → Policy	Real-time context
DF-05	Cloud	IVI ↔ Cloud	Telemetry, updates, analytics

VII. DISTINCTIVE AI/ML CONTRIBUTION

The distinctive element is a policy-first context layer. Instead of allowing each application to independently determine its presentation, the Context Fusion Engine publishes shared context to a Driver-Distraction Policy Engine. The policy engine is deterministic, so its decisions on which interactions are allowed can be reviewed and tested. The Personalization/Recommendation ML service can adapt recommendations, but it cannot override the policy boundary. This creates a clean separation between convenience intelligence and safety-oriented control logic.

VIII. SECURITY AND RELIABILITY

The architecture applies least-privilege access, gateway validation, projection-session isolation, encrypted cloud communication, structured telemetry and graceful degradation. Loss of cloud connectivity should not remove essential local IVI functions.

IX. TECHNOLOGY MAPPING

The design is platform-neutral at the service layer. It can be mapped to an Android Automotive-based IVI, an AUTOSAR Adaptive environment, or an OEM hybrid stack [1]–[4]. Service-oriented communication is used conceptually so that applications and services can evolve independently.

X. VALIDATION PLAN

Validation proceeds in the following order: (1) static architecture review; (2) interface-contract testing; (3) simulated CAN/GNSS events; (4) smartphone projection testing; (5) moving/parked HMI-policy testing; (6) network/cloud fault injection; and (7) performance, cybersecurity and usability validation.

XI. LIMITATIONS

This is a conceptual academic architecture, not a production automotive safety design or certification claim. Exact projection protocols, OEM vehicle interfaces, HMI rules and safety mechanisms vary by platform. Real deployment would require OEM-specific validation and applicable automotive development processes.

XII. CONCLUSION

DriveFusion IVI turns the required media–navigation–projection concept into a structured architecture with explicit interfaces, data contracts, security boundaries and an AI/ML-oriented context layer. Its central idea—deterministic policy first, personalization second—keeps convenience features separate from safety-oriented control. The design is suitable for an academic submission, viva demonstration and a future simulation prototype.

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